Quadratic Spline 3n×3n Solution Matrix

$$
\begin{vmatrix}
x_0^2 & x_0 & 1 & 0 & 0 & 0 & . & . & . \\
x_1^2 & x_1 & 1 & 0 & 0 & 0 & . & . & . \\
0 & 0 & 0 & x_1^2 & x_1 & 1 & . & . & . \\
. & . & . & . & . & . & . & . & . \\
. & . & . & . & . & 0 & x_n^2 & x_n & 1 \\
. & . & . & . & . & 0 & x_n^2 & x_n & 1 \\
2x_1 & 1 & 0 & -2x_1 & -1 & 0 & 0 & . & 0 \\
. & . & . & . & . & . & . & . & . \\
 & & & 2x_{n-1} & 1 & 0 & -2x_{n-1} & -1 & 0 \\
1 & 0 & 0 & 0 & & & & & 0
\end{vmatrix}
\times
\begin{vmatrix}
a_1 \\ b_1 \\ c_1 \\ . \\ . \\ . \\ . \\ a_n \\ b_n \\ c_n
\end{vmatrix}
=
\begin{vmatrix}
f(x_0) \\ f(x_1) \\ . \\ . \\ . \\ f(x_n) \\ 0 \\ . \\ 0 \\ 0
\end{vmatrix}
$$

Quadratic Spline Example

The upward velocity of a rocket is given as a function of time. Using quadratic splines

a) Find the velocity at t = 16 seconds

b) Find the velocity at t = 0 to t = 30 seconds

Table Velocity as a function of time

point	*t(s)*	*v(v) in m/s*
1	*0*	*0*
2	*10*	*227.04*
3	*15*	*362.78*
4	*20*	*517.35*
5	*22.5*	*602.97*
6	*30*	*901.67*

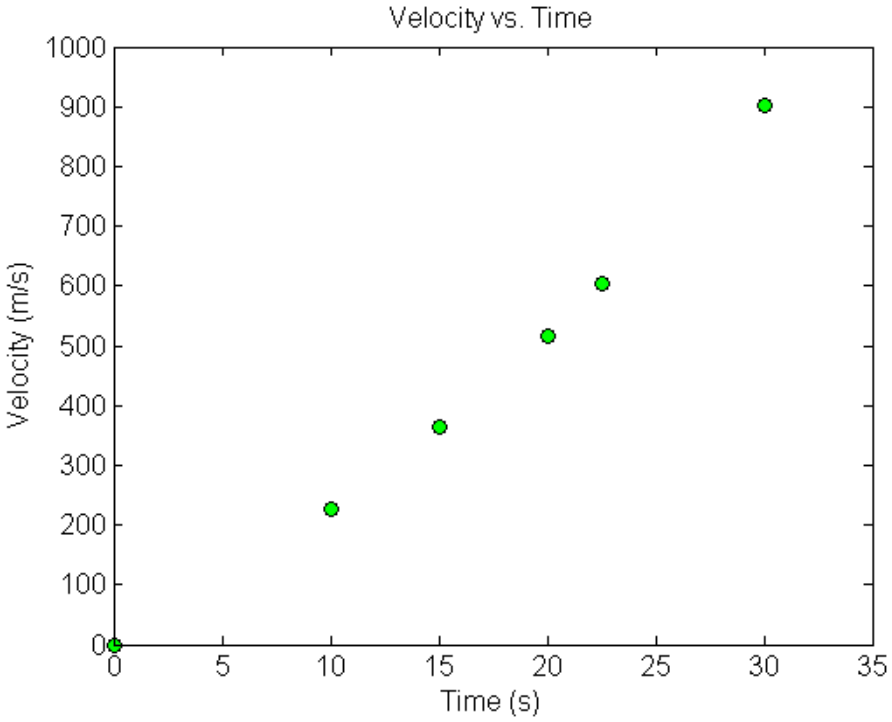


Figure. Velocity vs. time data for the rocket example

Final Set of Equations

$$
\begin{bmatrix}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 900 & 30 & 1 \\
20 & 1 & 0 & -20 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 30 & 1 & 0 & -30 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 40 & 1 & 0 & -40 & -1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 45 & 1 & 0 & -45 & -1 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\begin{bmatrix}
a_1 \\ b_1 \\ c_1 \\ a_2 \\ b_2 \\ c_2 \\ a_3 \\ b_3 \\ c_3 \\ a_4 \\ b_4 \\ c_4 \\ a_5 \\ b_5 \\ c_5
\end{bmatrix}
=
\begin{bmatrix}
0 \\ 227.04 \\ 227.04 \\ 362.78 \\ 362.78 \\ 517.35 \\ 517.35 \\ 602.97 \\ 602.97 \\ 901.67 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0
\end{bmatrix}
$$

Solving Set of Equations

1. Using Gauss Elimination method
2. Using LU Decomposition method
3. Other methods

Using Gauss Elimination method

Partial Pivoting

Gaussian Elimination with partial pivoting applies row swapping to normal Gaussian Elimination.

How?

At the beginning of the k_{th} step of forward elimination, find the maximum of

$$|a_{kk}|, |a_{k+1,k}|, \ldots\ldots\ldots\ldots, |a_{nk}|$$

If the maximum of the values is $|a_{pk}|$ In the p_{th} row, $k \le p \le n$,

then swap rows p and k. for $k = 1$ to $k = n-1$

Partial Pivoting: Example

$$\begin{bmatrix} 10 & 7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3.901 \\ 6 \end{bmatrix}$$

Solve using Gaussian Elimination with Partial Pivoting

Partial Pivoting: Example

<u>Step 1 (k = 1)</u>

- Examining the values of the first column |10|, |-3|, and |5| or 10, 3, and 5
- The largest absolute value is 10, which means, to follow the rules of Partial Pivoting, we have to swap row1 with row1.

$$\begin{bmatrix} 10 & 7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3.901 \\ 6 \end{bmatrix} \quad \Rightarrow \quad \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \\ 0 & 2.5 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 6.001 \\ 2.5 \end{bmatrix}$$

Partial Pivoting: Example

<u>Step 2 (k = 2)</u>

Examining the values of the first column |-0.001| and |2.5| or 0.0001 and 2.5

The largest absolute value is 2.5, so row 2 is swapped with row 3

$$\begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \\ 0 & 2.5 & 5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 6.001 \\ 2.5 \end{bmatrix} \Longrightarrow \begin{bmatrix} 10 & -7 & 0 \\ 0 & 2.5 & 5 \\ 0 & -0.001 & 6 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 2.5 \\ 6.001 \end{bmatrix}$$

Natural Cubic Spline Interpolation

- (Interpolated) Spline of degree of polynomial 3
- The domain of S is an interval [a, b].
- S, S', S'' are all continuous functions on [a, b].
- There are points t_i (the knots of S) such that $a = t_0 < t_1 < .. t_n = b$ and such that S is a polynomial of degree at most k on each subinterval $[t_i, t_{i+1}]$.

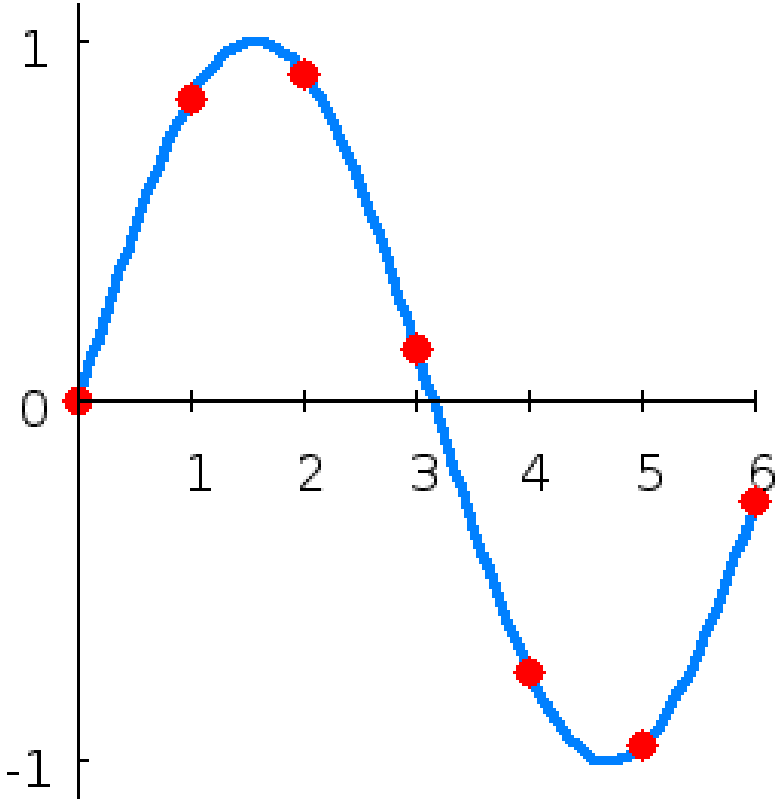
x	t_0	t_1	...	t_n
y	y_0	y_1	...	y_n

Natural Cubic Spline Interpolation

$$S(x) = \begin{cases} S_0(x), & x \in [x_0, x_1] \\ S_1(x), & x \in [x_1, x_2] \\ & \cdots \\ S_{n-1}(x), & x \in [x_{n-1}, x_n] \end{cases}$$

$S_i(x)$ is a cubic polynomial that will be used on the subinterval $[x_i, x_{i+1}]$.

Natural Cubic Spline Interpolation

- $S_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i$
  - 4 Coefficients with n subintervals = 4n equations
  - There are 4n-2 conditions
    - Interpolation conditions (3n)
    - Continuity conditions (n-2)

      and
    - Natural Conditions (2)
  - $S''(x_0) = 0$
  - $S''(x_n) = 0$